

Rootfinding

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Fixed Point Iteration Method

Theorem

Let

- 1 g be a continuous function on $[a, b]$
- 2 $g^{(\alpha)}$, $\alpha > 1$ continuous derivatives on (a, b)

Further, let $p \in (a, b)$ be a fixed point of g , i.e., $g(p) = p$. If

$$g'(p) = g''(p) = \cdots = g^{(\alpha-1)}(p) = 0, \text{ but } g^{(\alpha)}(p) \neq 0,$$

then there exists a $\delta > 0$ such that for any $x_0 \in [p - \delta, p + \delta]$, the sequence $x_{n+1} = g(x_n)$ converges to the fixed point p of order α with the asymptotic error constant

$$\lim_{n \rightarrow \infty} \frac{|e_{n+1}|}{|e_n|^\alpha} = \frac{|g^{(\alpha)}(p)|}{\alpha!}.$$

Note:

Using Mean Value Theorem,

$$\begin{aligned}|x_n - p| &= |g(x_{n-1}) - g(p)| = |g'(\xi)(x_{n-1} - p)| \\ &= |g'(\xi)||x_{n-1} - p| \\ e_n &= |g'(\xi)|e_{n-1} = |g'(\xi)|^2e_{n-2} = \cdots = |g'(\xi)|^ne_0\end{aligned}$$

Clearly, if $|g'(\xi)| < 1$, $|g'(\xi)|^n \rightarrow 0$ as $n \rightarrow \infty$. So, $x_n \rightarrow p$ as $n \rightarrow \infty$

Comparison of Fixed Point Method

- $f(x) = x^3 + x^2 - 3x - 3 = 0$
- Transform: $x = g_i(x)$, where

$$g_1(x) = \frac{x^3 + x^2 - 3}{3}, \quad g_2(x) = -1 + \frac{3x + 3}{x^2}$$

$$g_3(x) = \sqrt[3]{3 + 3x - x^2}, \quad g_4(x) = \sqrt{\frac{3 + 3x - x^2}{x}}$$

$$g_5(x) = x - \frac{x^3 + x^2 - 3x - 3}{3x^2 + 2x - 3}$$

- $g'_1(p) = 4.15$
- $g'_2(p) = -3.15$
- $g'_3(p) = -0.052$
- $g'_4(p) = -0.577$
- $g'_5(p) = 0, |g''_5(p)| = 1.3$

n	$g_1(x)$	(b) $g_2(x)$	(c) $g_3(x)$	(d) $g_4(x)$	(e) $g_5(x)$
0	+1.0	+1.0	+1.0	+1.0	+1.0
1	-.3333333333	5.0000000000	1.709975947	2.236067978	3.0000000000
2	-.9753086420	-.2800000000	1.733134316	1.451059202	2.2000000000
3	-.9921709716	26.5510204100	1.731994802	1.901682432	1.830150754
4	-.9974310264	-.8827544117	1.732053695	1.635808067	1.737795453
5	-.9991480696	-.5486245114	1.732050659	1.788336635	1.732072292
6	-.9997165068	3.4989256110	1.732050815	1.699764653	1.732050808
7	-.9999055558	.1024544340	1.732050807	1.750767137	1.732050808
8	-.9999685246	314.0796731		1.721269132	
9	-.9999895088	-.9904178717		1.738283891	
10	-.9999965030	-.9706946913		1.728454854	
11	-.9999988343	-.9066955739		1.734127847	
12	-.9999996114	-.6595130205		1.730851932	
13	-.9999998704	1.3484159170		1.732743080	
14	-.9999999568	2.8747932030		1.731651158	
15		.4065545020		1.732281557	
16		24.52938017		1.731917588	
17		-.8727117320		1.732127723	
18		-.4986188487		1.732006401	
19		5.0499512450		1.732076446	
20		-.2882970612		1.732036005	
25		Does not converge		1.732051758	

Newton's Method

- Start with a initial value x_0

1st Iteration:

- Draw a tangent at $(x_0, f(x_0))$
- Hence, equation of the tangent:

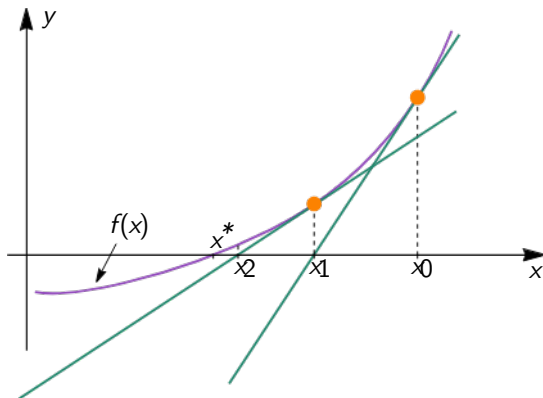
$$y - f(x_0) = f'(x_0)(x - x_0)$$

- Say, 1st iteration is obtained by the x-intercept $x = x_1$ of the tangent
- Substitute $x = x_1, y = 0$

$$-f(x_0) = f'(x_0)(x_1 - x_0)$$

which gives

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$



2nd Iteration:

- Draw a tangent at $(x_1, f(x_1))$
- Hence, equation of the tangent: $y - f(x_1) = f'(x_1)(x - x_1)$
- Say, 2nd iteration is obtained by the x-intercept $x = x_2$ of the tangent
- Substitute $x = x_2, y = 0$

$$-f(x_1) = f'(x_1)(x_2 - x_1)$$

which gives

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

(n+1)-th Iteration:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}, \quad n = 0, 1, 2, 3, \dots$$

Example

Find the root of the function $f(x) = x^3 + 2x^2 - 3x - 1$ on the interval $(1, 2)$ by applying Newton's method.

Solution: Compute the derivative of f : $f'(x) = 3x^2 + 4x - 3$. By Newton's method,

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}, \quad n = 0, 1, 2, 3, \dots$$

- **Initialization:** Say, $x_0 = 1$.
- **1st Iteration:** Compute $f(x_0) = f(1) = -1$, $f'(1) = 4$. Hence

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 1 - \frac{-1}{4} = 1.25$$

- **2nd Iteration:** Compute $f(x_1) = f(1.25) = 0.321$, $f'(1) = 6.6875$. Hence

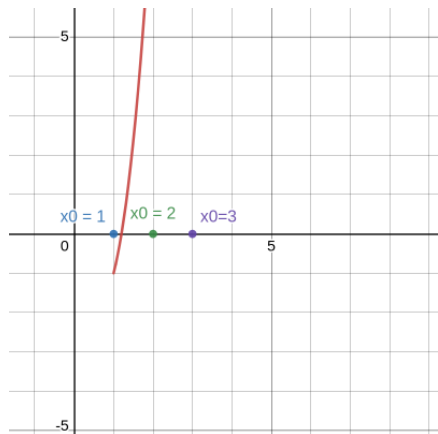
$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 1 - \frac{0.321}{6.6875} = 1.2009345794$$

- In a similar way,

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} = 1.1986958411$$

$$x_4 = x_3 - \frac{f(x_3)}{f'(x_3)} = 1.1986912435$$

Influence of initial guess, x_0



n	$p_0 = 1$	$p_0 = 2$	$p_0 = 3$
1	1.2500000000	1.4705882353	2.0277777778
2	1.2009345794	1.2471326788	1.4845011523
3	1.1986958411	1.2006987324	1.2514517238
4	1.1986912435	1.1986949265	1.2010586170
5		1.1986912435	1.1986963626
6			1.1986912435

Order of convergence

- If p is simple the roots of f , $f(p) = 0$ and $f'(p) \neq 0$
- In the view of Newton's method, we have $g(x) = x - \frac{f(x)}{f'(x)}$
- Clearly, $g(p) = p$
- Derivatives:

$$g'(x) = \frac{f(x)f''(x)}{[f'(x)]^2}$$

$$g''(x) = \frac{f''(x)}{f'(x)} + \frac{f'(x)f'''(x) - 2[f'(x)]^2}{[f'(x)]^3} f(x)$$

- At $x = p$, $g'(p) = 0$ and $g''(p) = \frac{f''(p)}{f'(p)} \neq 0$
- By the Theorem, Newton's method has order of convergence 2 with

$$\lim_{n \rightarrow \infty} \frac{|e_{n+1}|}{|e_n|^2} = \frac{|g''(p)|}{2!} = \frac{1}{2} \left| \frac{f''(p)}{f'(p)} \right|.$$

Newton's method is one of the most powerful tools for finding roots, but it has some disadvantages too:

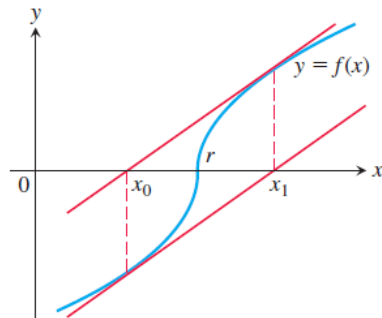
- Requires calculating derivatives of f . In many cases, it may not be easy for some complex form function.
- Newton's method will fail in the cases where the derivative f' is zero.

Newton's method does not always converge

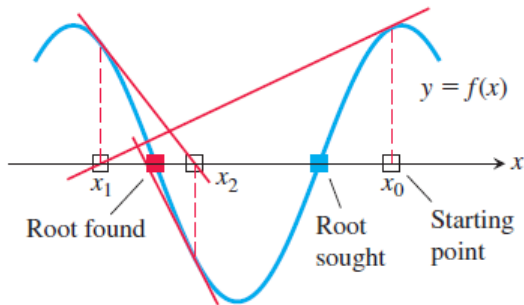
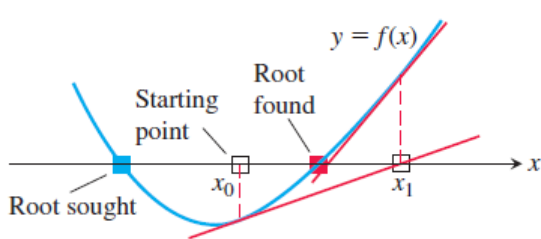
- Consider the function:

$$f(x) = \begin{cases} -\sqrt{r-x}, & x < r \\ \sqrt{x-r}, & x \geq r \end{cases}$$

- No amount of iteration brings us closer to the root than our first guess.



Newton's method converges to a root, it may not be the root you have in your mind



- If you start too far away, Newton's method may miss the root you want.

Newton's Method with roots of multiplicity > 1

- Issue: $f'(p) = 0$,
- Suppose $x = p$ is a root that appears m -times ($m \geq 2$), then f can be written as

$$f(x) = (x - p)^m q(x), \text{ with } q(p) \neq 0.$$

- Making Newton's method as a fixed point iteration function: $x_{n+1} = g(x_n)$, where

$$\begin{aligned} g(x) &= x - \frac{f(x)}{f'(x)} \\ &= x - \frac{(x - p)q(x)}{(x - p)q'(x) + mq(x)} \end{aligned}$$

- Its derivative

$$g'(x) = \frac{[m(m-1)q(x) + 2m(x-p)q'(x) + (x-p)^2q''(x)]q(x)}{[(x-p)q'(x) + mq(x)]^2}$$

- It is clear that $g(p) = p$ and $g'(p) = 1 - \frac{1}{m} = \frac{m-1}{m} \neq 0$
- Since rate of convergence of fixed point iteration method is linear, Newton's method provides only linear convergence for roots of multiplicity > 1 .
- Can we make it better ? rate of convergence is 2!!!
- Yes, we can do that by modifying $g(x) = [f(x)]^{\frac{1}{m}}$
- Applying Newton's method on g

$$x_{n+1} = x_n - \frac{[f(x_n)]^{\frac{1}{m}}}{\left\{ \frac{d}{dx} [f(x)]^{\frac{1}{m}} \right\}_{x=x_n}} = x_n - m \frac{f(x_n)}{f'(x_n)}$$

- How to find m ?
- If we know m , suppose $m = 2$, we will use $x_{n+1} = x_n - 2 \frac{f(x_n)}{f'(x_n)}$

- If m is unknown, then how use $x_{n+1} = x_n - m \frac{f(x_n)}{f'(x_n)}$?
- Assign a new function $u(x) = f(x)/f'(x)$, so

$$u(x) = \frac{(x-p)q(x)}{mq(x) + (x-p)q'(x)}, \quad u'(x) = \frac{f'(x)f'(x) - f(x)f''(x)}{[f'(x)]^2}$$

- If $f(x) = 0$ is having $x = p$ as a root, then $x = p$ is also root of $u(x)$
- Applying Newton's method on $u(x)$:

$$x_{n+1} = x_n - \frac{u(x_n)}{u'(x_n)} = x_n - \frac{f(x_n)f'(x_n)}{f'(x_n)f'(x_n) - f(x_n)f''(x_n)}$$

- Note that value of m is not used in the above formula (modified)

Example: Newton method vs Modified Newton Method

- Evaluate the multiple roots of $f(x) = x^3 - 5x^2 + 7x - 3$ with an initial guess $x_0 = 0$. (roots:1,1,3)

Newton method

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Newton Raphson

Iter	x_i	$\varepsilon_t\%$
0	0	100
1	0.4286	57
2	0.6857	31
3	0.83286	17
4	0.91332	8.7
5	0.95578	4.4
6	0.97766	2.2

Modified Newton method

$$x_{n+1} = x_n - \frac{f(x_n)f'(x_n)}{f'(x_n)f'(x_n) - f(x_n)f''(x_n)}$$

Modified Newton-Raphson

iter	x_i	$\varepsilon_t\%$
0	0	100
1	1.10526	11
2	1.00308	0.31
3	1.000002	0.00024

Secant Method

- Newton's method is very powerful tool, but it has a major weakness:

need to know f' at each iteration

- To address this, we substitute $f'(x_n)$ by $\frac{f(x_{n-1}) - f(x_n)}{x_{n-1} - x_n}$

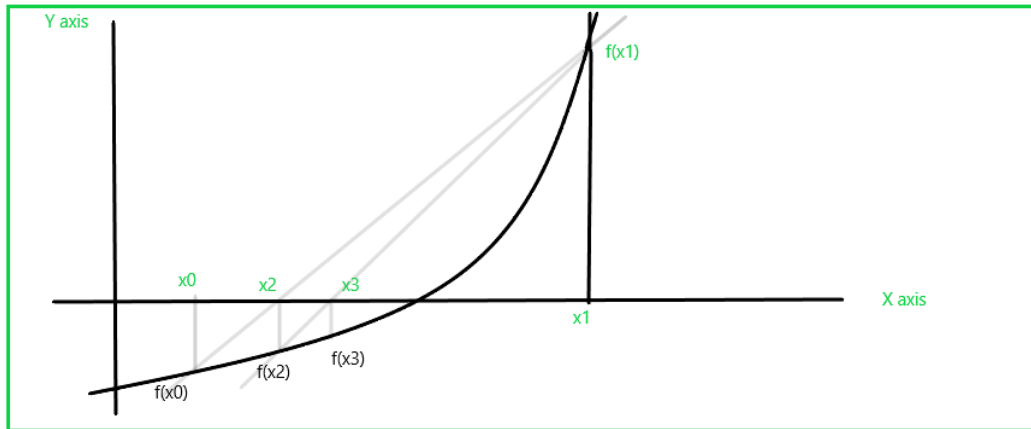
$$f'(x_n) = \lim_{x \rightarrow x_n} \frac{f(x_n) - f(x)}{x_n - x} \quad x_{n-1} \text{ is near to } x_n \rightarrow f'(x_n) \approx \frac{f(x_{n-1}) - f(x_n)}{x_{n-1} - x_n}$$

- Hence, Secant method is given by

$$x_{n+1} = x_n - f(x_n) \frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})}$$

- Note: Secant method needs two initial guess x_0 and x_1
- Only one function evaluation is required, where Newton's method needs two functions evaluation

Successive Secants



Example

Find the root of the function $f(x) = x^3 + 2x^2 - 3x - 1$ on the interval $(1, 2)$ by applying Secant method.

Solution: Taking initial guesses $x_0 = 2$ and $x_1 = 1$

1st Iteration:

$$x_2 = x_1 - f(x_1) \frac{x_1 - x_0}{f(x_1) - f(x_0)} = 1 - (-1) \frac{1 - 2}{-1 - 9} = 1.1$$

2nd Iteration:

$$x_3 = x_2 - f(x_2) \frac{x_2 - x_1}{f(x_2) - f(x_1)} = 1.1 - (-0.549) \frac{1.1 - 1}{-0.549 - (-1)} = 1.2217294900$$

Next 4 Iterations:

$$x_4 = 1.1964853266$$

$$x_5 = 1.198453684$$

$$x_6 = 1.1986913364$$

$$x_7 = 1.1986912435$$

Newton's method vs Secant method

- On the same accuracy, Secant method achieved in 6 iterations, but Newton's method achieved in only 4 iterations
- However, Secant method has six function evaluations, but Newton method has 8 function evaluations
- Thus, Secant method took more iterations than Newton's method to achieve a given level of accuracy, the Secant method used fewer function evaluations.

For another example, $f(x) = \tan(\pi x) - x - 6$ with level of accuracy of 9th decimal places

Method	Initial guess	Total Iterations	Total function evaluations
Newton's Method	$x_0 = 0.48$	5	10
Secant method	$x_0 = 4, x_1 = 0.48$	8	8

- Secant method converges at faster than a linear rate. (superlinear)
- Do not combine the secant formula and write it in the form

$$x_{n+1} = \frac{f(x_n)x_{n-1} - f(x_{n-1})x_n}{f(x_n) - f(x_{n-1})},$$

This has enormous loss of significance errors as compared with the earlier formulation.

Newton's method in the multivariable case

- n equations and n unknowns:

$$\left. \begin{aligned} f_1(x_1, x_2, \dots, x_n) &= 0, \\ f_2(x_1, x_2, \dots, x_n) &= 0, \\ &\vdots \\ f_n(x_1, x_2, \dots, x_n) &= 0. \end{aligned} \right\} \Rightarrow \mathbf{f}(\mathbf{x}) = \mathbf{0}, \text{ where } \mathbf{f} = \begin{bmatrix} f_1(\mathbf{x}) \\ f_2(\mathbf{x}) \\ \vdots \\ f_n(\mathbf{x}) \end{bmatrix}, \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

- Newton's method: $\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} - [\mathbf{J}(\mathbf{x}^{(k)})]^{-1} \mathbf{f}(\mathbf{x}^{(k)})$, $k = 0, 1, \dots$, with initial guess $\mathbf{x}^{(0)} \in \mathbb{R}^n$
- where $\mathbf{J}(\mathbf{x}^{(k)})$ is the Jacobian matrix at $\mathbf{x} = \mathbf{x}^{(k)}$ of size $n \times n$ such that

$$\mathbf{J}(\mathbf{x}^{(k)}) = \left[\frac{\partial f_i}{\partial x_j}(\mathbf{x}^{(k)}) \right]_{1 \leq i, j \leq n}$$

- $[\mathbf{J}(\mathbf{x}^{(k)})]^{-1}$ is the inverse of the Jacobian matrix

Example in 2D

Problem: Find a intersection points of the circle $x^2 + y^2 = 4$ and the hyperbola $4x^2 - y^2 = 4$.

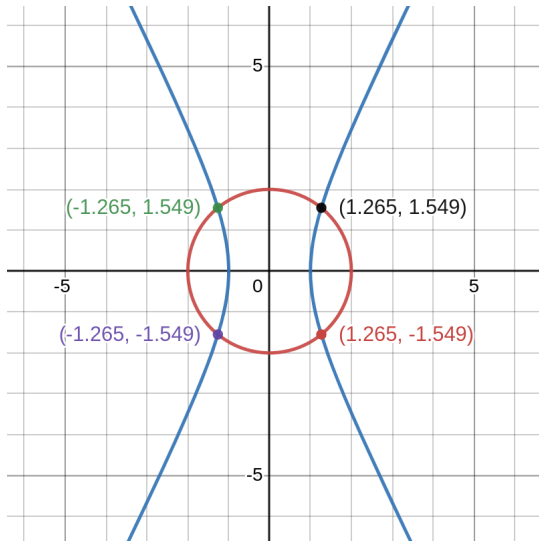
Solution. Equivalently, solutions to the system

$$f_1(x, y) = x^2 + y^2 - 4 = 0$$

$$f_2(x, y) = 4x^2 - y^2 - 4 = 0$$

So,

$$\mathbf{f}(\mathbf{x}) = \begin{bmatrix} f_1(\mathbf{x}) \\ f_2(\mathbf{x}) \end{bmatrix} = \begin{bmatrix} x^2 + y^2 - 4 \\ 4x^2 - y^2 - 4 \end{bmatrix}, \mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix}$$



Example contd1.

- Find the jacobian matrix

$$\mathbf{J}(\mathbf{x}) = \begin{bmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \end{bmatrix} = \begin{bmatrix} 2x & 2y \\ 8x & -2y \end{bmatrix}$$

- Find the inverse of the Jacobian matrix

$$[\mathbf{J}(\mathbf{x})]^{-1} = \frac{1}{-20xy} \begin{bmatrix} -2y & -2y \\ -8x & 2x \end{bmatrix} = \begin{bmatrix} \frac{1}{10x} & \frac{1}{10x} \\ \frac{2}{5y} & -\frac{1}{10y} \end{bmatrix}$$

- Compute $\mathbf{x}_{k+1} = \mathbf{x}_k - [\mathbf{J}(\mathbf{x}_k)]^{-1} \mathbf{f}(\mathbf{x}_k)$

$$\begin{bmatrix} x_{k+1} \\ y_{k+1} \end{bmatrix} = \begin{bmatrix} x_k \\ y_k \end{bmatrix} - \begin{bmatrix} \frac{1}{10x_k} & \frac{1}{10x_k} \\ \frac{2}{5y_k} & -\frac{1}{10y_k} \end{bmatrix} \begin{bmatrix} x_k^2 + y_k^2 - 4 \\ 4x_k^2 - y_k^2 - 4 \end{bmatrix} = \begin{bmatrix} x_k - \frac{5x_k^2 - 8}{10x_k} \\ y_k - \frac{5y_k^2 - 12}{10y_k} \end{bmatrix}$$

Example contd2.

- Consider the initial guess $\mathbf{x}_0 = \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$
- 1st Iteration:** when $k = 0$

$$\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} x_0 - \frac{5x_0^2 - 8}{10x_0} \\ y_0 - \frac{5y_0^2 - 12}{10y_0} \end{bmatrix} = \begin{bmatrix} 1 - \frac{5 * 1^2 - 8}{10 * 1} \\ 1 - \frac{5 * 1^2 - 12}{10 * 1} \end{bmatrix} = \begin{bmatrix} 1.3 \\ 1.7 \end{bmatrix}$$

- 2nd Iteration:** when $k = 1$

$$\begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \begin{bmatrix} x_1 - \frac{5x_1^2 - 8}{10x_1} \\ y_1 - \frac{5y_1^2 - 12}{10y_1} \end{bmatrix} = \begin{bmatrix} 1.3 - \frac{5 * (1.3)^2 - 8}{10 * (1.3)} \\ 1.7 - \frac{5 * (1.7)^2 - 12}{10 * (1.7)} \end{bmatrix} = \begin{bmatrix} 1.265385 \\ 1.555882 \end{bmatrix}$$

Example contd3.

- 3rd, 4th and 5th ... iterations:

$$\begin{bmatrix} x_3 \\ y_3 \end{bmatrix} = \begin{bmatrix} 1.26491115 \\ 1.54920772 \end{bmatrix},$$

$$\begin{bmatrix} x_4 \\ y_4 \end{bmatrix} = \begin{bmatrix} 1.26491106 \\ 1.54919334 \end{bmatrix},$$

$$\begin{bmatrix} x_5 \\ y_5 \end{bmatrix} = \begin{bmatrix} 1.26491106 \\ 1.54919334 \end{bmatrix}, \dots\dots$$

Problem Set

- 1 Use Newton's method to find solutions accurate within 10^{-5} on $[-2, -1]$: (a) $f(x) = x^2 - 2xe^{-x} + e^{-2x} = 0$ on $[0, 1]$, (b) $\cos(x + \sqrt{2}) + x(x/2 + \sqrt{2}) = 0$.
- 2 Use Secant method ($x_0 = 0.5, x_1 = \pi/4$) to find the solution $\cos x = x$, and compare the approximations with those given by Newton's method with $x_0 = \pi/4$.
- 3 Discuss the influence of x_0 and x_1 in Secant method.
- 4 Establish a comparison between Newton's method and the Secant method by considering the following function:

$$f(x) = x^3 + 2x^2 - 3x - 1, \quad \text{on } (-3, -2).$$

- 5 Let $f(x) = e^x - x - 1$. Show that f has a zero of multiplicity 2 at $x = 0$. Show that Newton's method with $x_0 = 1$ converges to this zero but not quadratically. Then show that modification of Newton's method improves the rate of convergence.
- 6 In example of Newton's 2D, take initial guesses $(1, -1), (-1, -1), (-1, 1)$ and $(0, 0)$, then discuss the convergence of roots.
- 7 Use Newton's method to find roots:
 $x^3 - 5x^2 + 2x - y + 13 = 0, x^3 + x^2 - 14x - y - 19 = 0$.